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| **Status** | Finished |
| **Started** | Sunday, 18 October 2026, 10:07 PM |
| **Completed** | Sunday, 18 October 2026, 10:10 PM |
| **Duration** | 3 mins 45 secs |
| **Grade** | **15.00** out of 15.00 (**100**%) |

Question **1**

Correct

Mark 1.00 out of 1.00

In which type of search is the list divided into smaller sublists during the search process?

a. Sequential search
b. Hash search
c. Binary search ✓
d. Linear search

Question **2**

Correct

Mark 1.00 out of 1.00

Which of the following statements about linear search is true?

a. Linear search can be applied to both sorted and unsorted lists. ✓
b. Linear search is more efficient than binary search on large lists.
c. Linear search requires the list to be sorted.
d. Linear search divides the list into halves.

Question **3**

Correct

Mark 1.00 out of 1.00

In linear search, how is the element searched?

a. By dividing the list into halves
b. By using a hash function
c. By comparing each element in the list sequentially ✓
d. By sorting the list first

Question **4**

Correct

Mark 1.00 out of 1.00

What is the key characteristic of binary search?

a. It works on unsorted lists
b. It compares elements sequentially
c. It always starts from the beginning of the list
d. It can be applied only if the list is sorted ✓

Question **5**

Correct

Mark 1.00 out of 1.00

During a linear search, what is the maximum number of comparisons needed to find an element in a list of size n?

a. log n
b. n/2
c. n-1
d. n ✓

Question **6**

Correct

Mark 1.00 out of 1.00

Which of the following is not a limitation of binary search algorithm?

a. Requirement of sorted array is expensive when a lot of insertion and deletions are needed
b. Must use a sorted array
c. There must be a mechanism to access middle element directly
d. Binary search algorithm is not efficient when the data elements more than 1500 ✓

Question **7**

Correct

Mark 1.00 out of 1.00

Which of the following is a limitation of binary search?

a. It can only be applied to large lists
b. It does not work with negative numbers
c. It requires the list to be sorted ✓
d. It is slower than linear search for small lists

Question **8**

Correct

Mark 1.00 out of 1.00

What is the time complexity of linear search in the worst case?

- a. O🥐 ✓
- b. O(log n)
- c. O(1)
- d. O(n log n)

Question **9**

Correct

Mark 1.00 out of 1.00

In binary search, what happens if the middle element does not match the target element?

- a. The search continues in the left or right sublist ✓
- b. The search continues from the beginning
- c. The list is sorted
- d. The search stops

Question **10**

Correct

Mark 1.00 out of 1.00

In binary search, if the target element is less than the middle element, where does the search continue?

- a. In the left sublist ✓
- b. At the beginning of the list
- c. In the entire list
- d. In the right sublist

Question **11**

Correct

Mark 1.00 out of 1.00

What is searching in the context of computer science?

- a. Deleting elements from a list
- b. Inserting elements into a list
- c. Determining whether an element is present in a list ✓
- d. Sorting elements in a list

Question **12**

Correct

Mark 1.00 out of 1.00

_____ search takes a sorted/ordered list and divides it in the middle.

- a. Binary ✓
- b. Hash
- c. Linear
- d. Both (1) & (3)

Question **13**

Correct

Mark 1.00 out of 1.00

In _____ checks the elements of a list, one at a time, without skipping any element.

- a. Binary search
- b. Both (1) & (3)
- c. Hash search
- d. Linear search ✓

Question **14**

Correct

Mark 1.00 out of 1.00

If the target element is greater than the middle element in binary search, where does the search continue?

- a. In the middle of the list
- b. In the right sublist ✓
- c. At the beginning of the list
- d. In the left sublist

Question **15**

Correct

Mark 1.00 out of 1.00

If a list contains 1000 elements, how many comparisons would a binary search typically make in the worst case?

- a. 500
- b. 1000
- c. 100
- d. 10 ✓